Team PJ

INTERLOCK BY TEAM PJ

Many agents. One codebase. Nothing breaks.

Interlock checks every edit an AI coding agent makes against what its teammates are changing, and stops the ones that would break their work.

INVESTOR DECK · NOVEMBER 2026 · TEAM-PJ-LAB.GITHUB.IO

VERDICTS · EXAMPLE

```
CLEAR agent-a auth.py
STOP agent-b checkout.py
WAIT agent-c cart.py
CLEAR agent-d README.md
```

THE PROBLEM

Each pull request passes. The app breaks.

AGENT A · auth.py

Changes a signature

```
def login(user, token):
```

CLEAR · its pull request passes

AGENT B · checkout.py

Calls the old one

```
session = login(user)
```

CLEAR · its pull request passes

STOP · AFTER MERGE

Both merge. checkout.py now calls login() without a token, and the app breaks. Neither agent could see the other's uncommitted work.

THE PROBLEM, AT SCALE

Agents already collide in the wild.

27.67%

of 107,026 pull requests opened by AI coding agents on GitHub conflicted with their target branch in a merge simulation.

AgenticFlict · Ogenrwot & Businge, 2026 · arXiv:2604.03551

33%

of 399 merges that version control reported as clean still broke the build or tests, in three open-source projects.

Brun, Holmes, Ernst & Notkin · ESEC/FSE 2011

WHY NOW

Agents went to work in a single year.

31% → 59%

of developers and tech professionals use AI agents at work, from the 2025 survey to April 2026. Daily use went from 14% to 37%.

Stack Overflow · 2025 Developer Survey and April 2026 Pulse Survey (n = 1,100)

16%

already run multiple overlapping or coordinated agents.

Stack Overflow · April 2026

85%

of 906 engineers use two or more AI tools, so one codebase gets agents from several vendors.

The Pragmatic Engineer · Jan–Feb 2026

OUR INSIGHT

Warning an agent isn’t enough.

WARNED BY SHEPHERD OR A NOTES FILE

12 of 12

runs still wrote the old call, though the agent had learned of the change.

STOP · breaks after merge

WITH INTERLOCK

12 of 12

calls fit the new signature. The old call was refused before it was written.

CLEAR · works after merge

Claude Sonnet 5.5 agents; a scripted teammate changed a function’s signature. So the edit itself has to be checked.

THE PRODUCT

Every edit is checked before it's written.

01 · AS IT WRITES

Claude Code edit hook

Each write is checked against what peers are changing.

02 · BEFORE A COMMIT

Git pre-commit hook

Covers agents that never ask permission.

03 · BEFORE A MERGE

GitHub check

Holds a pull request that depends on another branch's unmerged change.

EXAMPLE VERDICTS · NO LLM · RUNS LOCALLY OR SELF-HOSTED

STOP · agent-b · checkout.py calls login(user), which agent-a just changed

WAIT · agent-c · cart.py is queued until agent-a's edit lands

CLEAR · agent-d · README.md has no overlap

HOW IT DECIDES

Hard stops only where merges break.

STOP · ALWAYS REFUSED

- A call that doesn't fit a declared signature
- An edit to lines changed since the agent read them
- An edit to lines another agent is changing in the same checkout

CAUTION · REFUSED ONCE, THEN RECORDED

- An edit inside a function another agent holds
- An edit to another branch's lines

These often merge fine: holding whole functions would have blocked 1,709 real commit pairs that Git merges cleanly.

BEFORE AN AGENT STOPS Interlock tests its work merged with its peers', including peers on other machines.

REPLAY OF 5,352 REAL COMMIT PAIRS · HTTPX, CLICK, REQUESTS, ZOD, KY

HEAD TO HEAD · LATEST ROUND

60 of 60 runs finished correctly.



MEDIAN TIME TO FINISH

54 s Interlock · 71 s no tool · 100 s Shepherd

Fastest condition in each of the last 10 rounds, at about the same API cost as no tool.

SETUP

3–5 Claude Code agents on Claude Haiku 5.5, each on its own simulated machine. Correct means all work merged and passed tests the agents never saw.

Limits: we design the scenarios and commit them before each run; small Python codebases (74 lines in this round); machines simulated on one computer. On Claude Sonnet 5.5, no tool also scored 59/60.

PRECISION AND RECALL

Strict on conflicts. Quiet on clean work.

651 / 651

interface conflicts caught on held-out codebases written by a separate AI agent that never read our source.

0 of 3,535

clean commit pairs refused, replaying 5,352 real pairs from httpx, click, requests, zod and ky.

2 of 1,817

pairs counted as conflicts that the line rule missed (an approximate count). We publish the misses too.

3–6 ms

to check one write against peers' uncommitted lines (median, on a 2,000-line file).

TRACTION

Early, real, and logged.

42
releases, October 16–30

1,000+
automated tests passing

164
decisions logged in one workday

FIRST PROFESSIONAL USER

A working web developer outside our school ran up to 4 Claude Code agents on one repository on Oct 29. Interlock stopped 28 edits and queued 14.

He brought in three colleagues and says he'd keep using it once it's paid, at about $25 per user or $200 per team a month (non-binding).

STUDENT DEVELOPERS

21 of 25 members of Bugeon's high school AI club used Interlock on Oct 20.

10–12 have kept using it since, by the club president's written account (no usage logs).

MARKET

≈$1.2B a year, counted bottom-up.

36.5M
professional developers worldwide
SlashData, early 2025

×

59%
use AI agents at work
Stack Overflow, Apr 2026

×

16%
run multiple overlapping or coordinated agents
Stack Overflow, Apr 2026

=

≈3.4M
developers running several agents today

≈$1.2B
a year today: 3.4M developers at $29 per developer per month.

≈$12.7B
a year if every professional developer worked this way, at the same price.

OUR ESTIMATE · SOURCES AS LISTED

BUSINESS MODEL

We charge for coordination, not tokens.

ENGINEERING TEAMS · FIRST

$29

per developer per month

We'll test this price with professional teams first.

INDIVIDUAL DEVELOPERS

$19

per month

ENTERPRISE

Annual

contracts

Self-hosting, central controls and security features.

≈ $35M ARR is what 10,000 teams of 10 developers each would pay at $29 a month.

PRICING TO BE VALIDATED · FIRST USER'S NON-BINDING QUOTE: ~$25/USER, $200/TEAM

COMPETITION

Others warn. Interlock checks the edit.

Approach	What happens
Worktree per agent	Dependencies stay hidden until merge.
File locks	Blocked all 3,535 clean commit pairs we replayed.
Warn-only tools (Shepherd, Foremerge)	Warned agents still wrote the old call in 12 of 12 runs.
One vendor's agent teams	Coordinate one developer's session; the docs say to give each agent different files.
Interlock	Checks every edit, commit and pull request, across machines and vendors.

In one round against four other tools: Interlock 60/60, Shepherd 42, no tool 40, Foremerge 38, mcp_agent_mail 36, clash 35 (0 unattended).

A vendor checks its own agents; Interlock checks everyone's.

TEAM

Friends since 2019. Building since 2023.

Woojin Jung

CEO · LEADS PRODUCT

- Built Interlock V1.0–V2.0, the design it still enforces
- Special Prize, Korea's 71st National Science Exhibition (2025)
- Deputy Lead of ARION, DGIST's rocketry club; built its simulator RSIM
- Student at DGIST, Daegu

Bugeon Park

CTO · LEADS ENGINEERING

- Has led every Interlock release since V2.0
- Part-time CTO at Votavis: 10 developers to a finished MVP in four months
- Trained jamo2hangul, a 53M-parameter Transformer, from scratch
- 3rd of 198 teams, Korea's national high-school SW challenge (2026)

Met at the same elementary school; first built together an automated crypto trading system. Equity 50/50; Woojin makes the final call.

HOW WE WORK

Proof before promises.

01 · COMMITTED FIRST

Scenarios and pass criteria are committed before every run.

02 · LOSSES ONLINE TOO

Every benchmark round goes online, including the ones Interlock lost.

03 · LOSE, THEN FIX

When a rival wins on the numbers, we change the code, not the test.

FIRST FULL HEAD-TO-HEAD

Shepherd 68%, Interlock 37%.

LATEST ROUND

Interlock 60/60,
Shepherd 37/60.

IN BETWEEN

42 releases in 15 days.

ROADMAP

Next: many agents, one working system.

NOW · LIVE	Edits, commits and pull requests checked; Claude Code hooks live.
DEC 2026 – JAN 2027	The public beta opens.
NEXT	Codex and Cursor hooks on live agents; full JS/TS signature checks.
THEN	Task assignment, execution order and recovery across agents.

WHAT WE'RE LOOKING FOR

A partner to back us full-time in San Francisco. If accepted, both founders go full-time.

GET IN TOUCH

Put Interlock between your agents.

Email us for the manual and a beta access token. The public beta opens December 2026 – January 2027.

pj.southkorea@gmail.com

team-pj-lab.github.io

team-pj-lab.github.io/interlock/experiments

WOOJIN JUNG, CEO · BUGEON PARK, CTO